
Trust-Calibrated Multi-Agent Scientific Deliberation for Mitigating Sycophantic Consensus in LLM Reasoning

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Submitted in partial fulfilment of the requirements
of the degree of Bachelor of Science in Computer Science and Engineering

2026



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Abstract

Large language models answer many scientific questions well, yet they still hallucinate and follow broken lines of reasoning. Multi-agent debate is one answer to this. Several models argue the same question before a final answer is chosen. The weak point is the final decision. Most debate systems count votes, so the majority wins. A confident but wrong majority can drag a correct minority agent onto the wrong side, and the group settles on a wrong answer. That is the sycophantic consensus problem. No current method checks an agent’s claims against outside scientific evidence before that agent gains influence. This work changes how the debate is settled. The aim is to weight each agent by evidence instead of headcount. Every response is split into small factual claims. For each claim, the system retrieves relevant scientific literature and checks whether the evidence supports it or contradicts it. Each agent carries a trust score that rises and falls as these checks build up across the rounds. Unsupported claims lose influence, and the score stays bounded so no agent dominates the discussion. The reward is a correct minority that survives. Its influence no longer depends on the number of voices behind it. The group is less likely to agree on a wrong answer too early. The method needs no fine-tuning. The framework is expected to reduce hallucinations and keep reasoning tied to verifiable sources. Comparisons cover majority voting and other baselines. Accuracy, consensus-collapse rate, minority-preservation rate, and evidence-calibration rate are all measured, and the final answer carries the citations behind it.

Acknowledgements

Completing this project took more than the six of us. We are grateful to everyone who helped along the way.

Dr. Mohammad Nurul Huda supervised the work. His feedback kept the scope realistic and pushed the technical depth in the right places, and the project is better for it.

Dr. Ohidujjaman, the FYDP course teacher, made the process itself clear to us. Her deadlines and instructions shaped how the team worked across the trimesters.

Friends and peers contributed in smaller but steady ways, from proofreading chapters to helping us reason through stubborn bugs.

Our families carried the quiet part of the load, and we thank them for the patience and support that kept us going.

Publication List

The publications connected to this research are listed below. One journal article is currently under review, and further submissions are planned once the evaluation results are ready.

Journal Articles

1. Trust-Calibrated Multi-Agent Scientific Deliberation for Mitigating Sycophantic Consensus in LLM Reasoning (In preparation)

Conference Papers

- 1.

Additional Publications

Following is the list of relevant publications published in the course of the research that is not included in the thesis:

- 1.

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Chapter 1

Introduction

The chapter introduces the study. It presents the background and problem, the motivation behind the work, the objectives, a short summary of the methodology, the expected outcome, and the structure of the rest of the report.

1.1 Project Overview

Large language models now handle many reasoning tasks, from scientific question answering to tutoring and decision support. A single model can still be confidently wrong. It may state an incorrect answer in fluent and assured language, which makes the mistake hard to catch. Multi-agent debate was proposed to reduce this problem. Several agents answer the same question, read each other’s reasoning, and revise their positions over a few rounds before a final answer is selected [1]. Chain-of-thought prompting gives each agent a way to expose its intermediate steps [2], which makes the reasoning inside a debate easier to inspect. The real difficulty is the final step, where the separate answers are combined. Most debate systems settle disagreements by majority vote. Each agent receives the same weight, so the most common answer wins. This rule treats popularity as a stand-in for correctness. In scientific reasoning, a better answer should be backed by evidence rather than repeated by more agents. The study examines a trust-calibrated multi-agent deliberation framework that adjusts each agent’s influence during the debate based on how well its claims match retrieved external evidence. The final answer then depends on evidence-grounded trust rather than a raw headcount.

1.2 Motivation

Majority voting can fail in a way that is easy to miss. A debate may look like it reached agreement while it has quietly dropped a correct answer held by one agent. Recent studies of multi-agent debate report that agents often copy or switch answers under social pressure instead of reasoning on their own, and that the correct answer is present in the discussion but ignored in more than 20 percent of wrong-answer cases [3]. Work on model behaviour

points to a related tendency, where models trained to be agreeable grow more sycophantic and less accurate [4]. A confident majority pushes a correct minority agent to give up its answer, and majority voting then does more than pick the wrong result. It hides the failure behind an appearance of consensus. This matters most for scientific questions, where a wrong but confident consensus can mislead a user into trusting an answer that deserved a closer check. Each existing method attacks only part of the problem. iMAD decides when a debate is worth running at all [5]. DebUnc scales agent influence by self-reported confidence [6]. Mixture-of-Agents folds all outputs together through static aggregation [7]. None of these ties an agent’s influence to whether its claims survive a check against external evidence while the debate is still running. That connection is exactly what this study adds.

1.3 Objectives

The main objective is to reduce sycophantic consensus in multi-agent LLM debate. Plain majority voting gets replaced with evidence-grounded trust calibration. Reaching that goal needs a few things. The first is a clear picture of how a confident wrong majority pressures a correct minority agent into changing its answer [3, 8]. The second is a bounded trust score. It rises when an agent’s claims are supported by external evidence and falls when they are contradicted, and it is applied before the final answer is fixed. The study also compares the method against standard majority voting and other debate and aggregation baselines [9, 10], using answer accuracy together with consensus-collapse, minority-preservation, and evidence-calibration metrics.

1.4 Methodology

The study follows an experimental design. A baseline multi-agent debate system is built first. Several heterogeneous LLM agents answer the same scientific question, read each other’s reasoning, and revise their positions across rounds [1]. The baseline keeps majority voting so the usual failure mode can be observed and measured [9]. Each agent’s reasoning is then split into smaller factual claims. The claims are checked against scientific evidence retrieved from external sources such as academic papers and reference databases [11]. Contradictions between a claim and the retrieved evidence are detected and scored [12]. This builds on the idea that external tools can verify and correct model output [13]. The verdicts feed into trust updates, following the line of work that uses per-agent confidence to steer debate [6]. Supported claims raise an agent’s trust. Contradicted claims lower it. Uncertain claims are handled conservatively. The trust score stays bounded, so no agent gains unlimited influence or drops out of the debate completely. The final answer comes from trust-weighted aggregation rather than a simple count. The framework is tested on scientific reasoning datasets such as GPQA and MMLU-Pro. The evaluation looks at three things: whether the mechanism lowers consensus collapse, whether correct minority answers survive, and whether overall accuracy holds, all without any model fine-tuning.

Figure 1.1 shows the end-to-end flow of the framework.

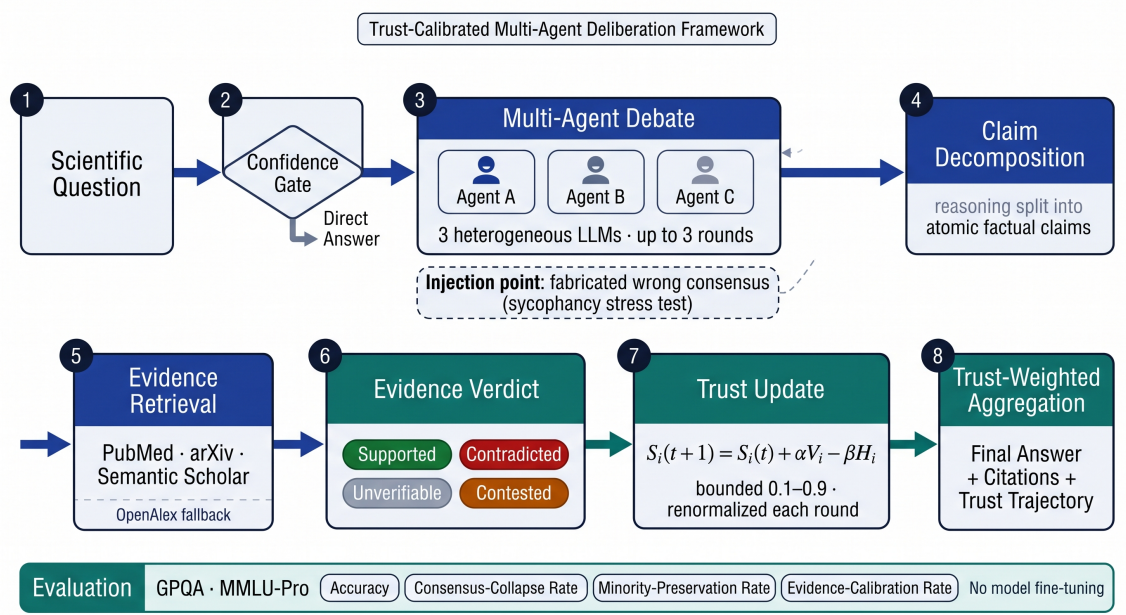


Figure 1.1: End-to-end flow of the trust-calibrated multi-agent deliberation framework: debate, claim-level evidence verification, per-round trust updates, and trust-weighted aggregation.

1.5 Project Outcome

The main outcome is a reproducible framework for trust-calibrated multi-agent scientific deliberation. Retrieved evidence updates agent influence while the debate runs, and the final answer rests on evidence-weighted trust instead of a majority vote. The study also produces an evaluation protocol. It detects sycophantic consensus [3], tracks how often correct minority answers survive [8], and measures the method against established baselines. The hope is that dynamic trust calibration makes multi-agent debate more dependable on scientific questions. Where the method fails to remove every wrong answer, it should still make the decision more transparent by recording which agents the evidence supported and which it contradicted.

1.6 Organization of the Report

The rest of the report is arranged as follows. Chapter 2 covers the background, the related work on multi-agent debate and sycophancy, and the gap addressed by the study. Chapter 3 presents the requirement analysis, the detailed methodology, the system design, and the project plan. Chapter 4 describes the implementation setup, the testing and evaluation procedure, and the results with discussion. Chapter 5 discusses the relevant standards, the

impact of the work, sustainability, and the complex engineering problem analysis. Chapter 6 concludes the report, notes its limitations, and gives recommendations for future work.

Chapter 2

Background

The chapter introduces the concepts needed to understand the system described in the report and reviews research on multi-agent debate, sycophancy, and agent influence. It then explains the gap addressed by the study.

2.1 Preliminaries

Large language models (LLMs) generate answers one token at a time, but their outputs can still contain unsupported or incorrect reasoning. Fluent wording does not guarantee that an answer is correct [14]. Chain-of-thought prompting asks a model to expose intermediate reasoning steps before giving a final answer, which makes the reasoning easier to inspect and gives later components more information to evaluate [2]. These steps are useful evidence about how an answer was formed, but they should not be treated as proof of correctness by themselves. Multi-agent debate takes this further. Several agents solve the same problem on their own, then see the other answers and explanations, discuss the differences, and revise their positions across a fixed number of rounds. Only after the discussion is a final answer chosen [1]. Independent attempts help because different agents expose different errors, and one agent can question another agent’s reasoning. How the answers get combined still matters. Majority voting is the common choice: every agent carries the same weight and the most frequent answer wins. The vote then strengthens a repeated mistake instead of correcting it [10]. Sycophantic consensus describes the case where an agent copies a wrong position or gives up a correct one simply because the wrong position has more support [3]. Confidence can come from the agent itself or from its token distribution, but neither source verifies the underlying claim [6]. DebUnc makes this point by comparing deployable uncertainty measures against a diagnostic Ground Truth measure that already knows the answer [6]. The framework instead builds trust from retrieved scientific evidence.

2.2 Literature Review

The review sorts the literature into five areas: debate construction, external verification, sycophancy, uncertainty, and debate evaluation. The main comparison sits between methods that decide when to debate, methods that change agent interaction, and the method that changes agent influence using retrieved evidence. Surveys of LLM-based multi-agent systems map the design space across workflow, infrastructure, and open challenges [15]. Recent reasoning-oriented LLMs show that better reasoning needs more than a bigger model. Research has moved toward structured reasoning, multi-agent collaboration, and external verification to make generated responses more reliable [14]. Multi-agent debate is among the most studied of these directions, because agents can challenge each other’s reasoning before a final answer is produced [1]. The following review goes through representative studies in these areas and points to the research gap the framework addresses.

2.2.1 Similar Applications

The system is not a conventional web or mobile application. Instead, it is an experimental deliberation layer that can be integrated into scientific question-answering systems, research assistants, evidence-based decision-support tools, or other applications that require reliable reasoning. Rather than relying on a single model response, these systems generate multiple candidate solutions, allow the candidates to interact, and aggregate the results before returning a final answer to the user. Du et al. demonstrated that structured debate among multiple agents can improve factuality and reasoning compared with a single-agent approach [1]. Wang et al. further showed that combining the outputs of several models through layered aggregation can improve overall model capability without requiring explicit debate [7]. Existing multi-agent applications differ mainly in how they coordinate agents and combine their outputs. Some systems focus on structured debate to encourage reasoning refinement between agents [1]. Other systems organize multiple agents into hierarchical aggregation pipelines where later models refine or combine earlier responses [7]. Zhou et al. described a self-organizing setup where the task itself decides which reasoning agents take part, instead of a fixed workflow [16]. Kushwaha and Madhavan built a retrieval-augmented system that spots contradictions between retrieved documents and ranks evidence by source credibility before producing a final response [12]. Both directions show that spreading reasoning across specialized agents helps robustness and factual consistency. Dialogue design and scale also matter. Ueda et al. studied how multi-agent LLM dialogues should be structured for research ideation [17]. Wang et al. looked at how well LLM-based multi-agent systems scale for scientific code generation [18]. Their aggregation stays fixed once the discussion starts. The proposed framework keeps the same goal, reliable answers, but lets agent influence change during the discussion. Evidence verification results are recorded after every round, and each agent’s trust score is updated from claim-level scientific evidence. That way a well-supported minority opinion can survive, and an unsupported majority consensus carries less weight.

2.2.2 Related Research

Wei et al. (2022) showed that chain-of-thought prompting improves reasoning in large language models [2]. Du et al. (2024) built on that with multi-agent debate, where agents answer a question, look at peer reasoning, and revise their positions [1]. CRITIC takes a different route [13]. It uses external tools to check and improve a model’s response, so claims can be verified, but only one model works on the question at a time. Mixture-of-Agents is layered instead of conversational [7]. Several proposer models produce answers and a later stage combines them, which shows what multiple LLMs can do, yet the combination is static and carries no evidence-based trust value. iMAD attacks cost rather than correctness [5]. Fan, Yoon, and Ji extract 41 features from a structured self-critique and let a classifier decide whether a debate should start at all. They report token savings of up to 92 percent against GroupDebate and accuracy gains of up to 13.5 percentage points over a single-agent baseline. Once the debate begins, agent influence stays unchanged. Pitre, Ramakrishnan, and Wang studied sycophancy inside debate [3]. In their experiments the correct answer was present but ignored in more than 20 percent of wrong-answer cases. CONSENSAGENT detects stalling and answer copying, then rewrites the task prompt before the next debate stage. Measured sycophancy drops, but the final aggregation still leans on self-reported confidence and consistency. DebUnc instead shares uncertainty between agents [6]. Uncertainty reaches peers through prompts or attention scaling, yet the deployable measures give small gains. A diagnostic Ground Truth signal, which knows the correct answer, gives a much larger gain, and the gap between the two exposes the real bottleneck: the quality of the trust signal. Kushwaha and Madhavan described an eight-agent MAS-RAG system that detects contradictions among retrieved documents and ranks evidence by credibility [12]. Source credibility here is fixed by a human curator and never updated while deliberation runs. Other studies look at the social side of debate. Cau et al. found that selective persuasion can steer opinion convergence [19]. Yeow et al. compared decision architectures and found that feedback between agents can spread early errors [20]. Wang et al. showed that a single LLM with strong in-context examples can sometimes match multi-agent discussion, and they list judge errors and peer conformity as weak points [10]. Choi, Zhu, and Li separated debate communication from voting and argued that majority voting explains much of the usual gain [9]. FREE-MAD goes further and drops final consensus altogether, using anti-conformity prompts and trajectory scoring [21]. ReSo selects agents through reward-driven dynamic task graphs [16]. These works tune coordination and agent selection, but none verifies scientific claims before changing agent influence. Sycophancy also appears at the model level. Ibrahim et al. reported that tuning models to sound warm and agreeable reduces factual accuracy and increases sycophancy [4]. Chen et al. attacked the same problem from the training side with self-augmented preference alignment [22]. Pitre et al. proposed process-level diagnostics for engagement, responsiveness, influence asymmetry, balance, stability, and agent utility, useful when ground truth is unavailable [23]. Final agreement alone, their

work suggests, is not enough to judge a debate. Across all of these studies, no method continuously updates an agent’s influence from claim-level support in retrieved scientific evidence. That is the gap the framework closes with a bounded trust score.

2.3 Gap Analysis

The reviewed literature reveals three remaining needs in multi-agent debate systems. First, a system should determine whether the additional computational cost of the debate is justified. Fan et al. addressed this through iMAD by predicting whether a debate is necessary before it begins [5]. Second, a system should detect unsupported agreement caused by conformity or sycophancy. CONSENSAGENT mitigates this problem through prompt optimization during debate [3]. Third, a system should determine which agents deserve greater influence when disagreement remains. DebUnc introduces uncertainty-based weighting, but its trust signals rely on internal model uncertainty rather than independently verifiable evidence [6]. Previous studies also show that agreement does not necessarily indicate correctness. Cau et al. found that selective persuasion influences opinion convergence [19]. Wang et al. identified peer conformity as a limitation of multi-agent debate [10]. Existing retrieval-based systems verify retrieved information or improve aggregation, but they do not continuously update an agent’s influence according to verified scientific evidence during debate [13]. Dynamic coordination methods also focus on agent selection rather than evidence-driven trust updates [16]. The remaining gap is therefore an evidence-grounded mechanism that continuously updates agent influence during an ongoing debate. The proposed framework addresses this gap by decomposing an agent’s response into factual claims, retrieving relevant scientific passages, and classifying each claim as supported, contradicted, or unverifiable. These verification results update a bounded trust score after every debate round, allowing well-supported minority agents to retain influence while reducing the impact of unsupported majority consensus. The framework is evaluated using consensus-collapse rate, minority-preservation rate, and evidence-calibration rate alongside conventional answer accuracy. Retrieval sparsity, imperfect claim decomposition, and correlated reasoning errors between agents remain limitations of the proposed approach.

2.4 Summary

The chapter introduced LLM reasoning, chain-of-thought prompting, multi-agent debate, and sycophantic consensus. The reviewed studies cover debate efficiency, prompt correction, uncertainty weighting, external verification, process evaluation, and agent coordination. They do not continuously update agent influence from claim-level scientific evidence. The next chapter uses this gap to define the system requirements, detailed methodology, and project plan.

Chapter 3

Project Design

This chapter defines the system requirements, presents the detailed methodology and the design decisions behind the framework, and lays out the project plan with the task allocation for the team. The first section records the functional and nonfunctional requirements together with the context diagram, the level-one data flow diagram, and the interface design.

3.1 Requirement Analysis

Requirement analysis fixes what the framework must do before any code is written. The system is an experimental deliberation layer rather than a conventional application, so the requirements are framed around the research pipeline. A question enters, agents debate it over a fixed number of rounds, the claims are checked against external evidence, and a trust-weighted answer comes out. Multi-agent debate was chosen because it has been shown to improve factuality over single-model answers [1]. The trust mechanism exists because a confident wrong majority can push a correct minority agent into abandoning its answer [3]. The subsections below define the requirements, the system boundary, the data flows, and the interface through which the framework is operated.

3.1.1 Functional and Nonfunctional Requirements

The functional requirements describe the services the framework must provide. The system accepts a scientific question from a user or the evaluation harness and keeps the ground truth hidden from all agents. For every question the confidence gate decides whether a debate is needed, sends easy questions to a direct answer, and routes the remaining questions into the debate pipeline. The orchestration process runs three heterogeneous LLM agents over at most three revision rounds, with a retry cap of three, and sequences the rounds with a state machine. Each agent’s reasoning is split into atomic factual claims tagged to their source agent, with a fallback extraction pass when structured tagging fails. The retrieval subsystem searches each agent’s claims in a different database, PubMed for the first agent, arXiv for the second, and Semantic Scholar for the third, with

OpenAlex as a fallback when a source is unavailable. Retrieved passages are reranked with a cross-encoder and every claim receives a verdict of supported, contradicted, unverifiable, or contested. After every round the trust score of each agent is updated with $S_i(t+1) = S_i(t) + \alpha V_i - \beta H_i$, then softmax is applied, the values are clamped between 0.1 and 0.9, and the vector is renormalized so that no agent is silenced or dominant. The system also supports an injection point where a fabricated wrong consensus can be inserted between rounds, so that sycophantic collapse can be measured under controlled pressure. The final answer is selected by weighted aggregation of trust and position at the last round rather than by majority vote, and the result package contains the answer, the supporting citations, the full trust trajectory, and each agent’s reasoning trace. The evaluation harness computes answer accuracy, consensus-collapse rate, minority-preservation rate, and evidence-calibration rate, and compares the framework with majority voting and with the MoA [7], iMAD [5], and DebUnc [6] baselines. Finally, the system recovers unparseable model output with a fallback extraction pass, retries a timeout once before marking the round inconclusive, and falls back to cached evidence or OpenAlex when a retrieval API is down.

The nonfunctional requirements describe the qualities of the system rather than its functions. Latency is the first constraint. The debate must respect the round cap of three and the retry cap of three, so latency stays bounded on the target hardware. A failure in one component does not block the whole debate. Every failure path degrades locally and keeps the pipeline moving. Portability matters too. Closed APIs and locally served models both work through the OpenAI-compatible vLLM interface. The pipeline runs unchanged on the development models (Qwen3.5-9B, Gemma 4 12B, Ministral-3-14B) and on the final models (Qwen3.6-27B, Gemma 4 26B, Mistral Small 3.2 24B). Inference fits within a single RTX PRO 6000 Blackwell 96GB rental at about USD 0.70 to 1.90 per hour, with no fine-tuning. Every run is seeded and every intermediate result is logged, so any experiment can be repeated and checked. The output records which agent was supported or contradicted by the evidence, leaving a decision trail that can be audited. The ground truth stays hidden from the agents, and the final output cites the evidence behind the answer.

3.1.2 Context Diagram

The context diagram in Figure 3.1 shows the system boundary and the external actors it exchanges data with. The framework sits in the middle of the diagram. It receives questions from the user or the evaluation harness, retrieves scientific passages from three external databases, and calls off-the-shelf language models through the vLLM server running on a rented RTX PRO 6000 Blackwell GPU. The evaluation harness receives the scores produced by the framework, and the user receives the final result package with the answer, the evidence, and the trust trajectory.

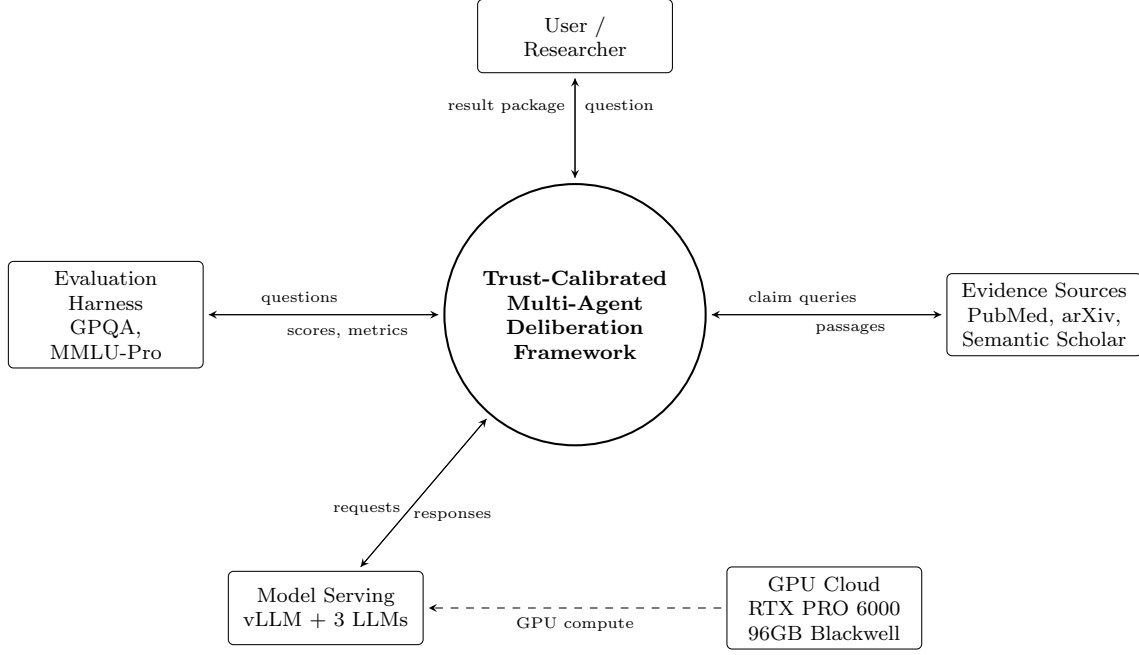


Figure 3.1: Context diagram of the trust-calibrated multi-agent deliberation framework.

3.1.3 Data Flow Diagram Level 1

The level-one data flow diagram in Figure 3.2 decomposes the framework into seven processes. A question enters through the intake and gate process, which decides between a direct answer and a debate. The orchestration process runs the revision rounds, sends inference requests to the external models, and reads and writes the debate state. Each position is decomposed into atomic claims and verified against the external sources. The verdicts drive the trust update, which writes the new trust values back into the debate state. At the last round the weighted aggregation process selects the answer, and the packaging process logs the result and returns it to the user. Three data stores hold the debate state, the evidence and verdict log, and the results log.

The diagram reads from left to right and top to bottom. The evaluation harness submits a question, the confidence gate decides whether the debate is worth running, and the orchestration process asks the model server for the agent positions. Each position is decomposed into atomic claims, the retrieval process verifies them against the external sources, and the verdicts are stored in the evidence log before they drive the trust update. The updated trust is written back into the debate state, so the next round starts from the corrected influence values. At the last round the aggregation process selects the answer, the packaging process logs it in the results log, and the result package with the answer and the citations returns to the user.

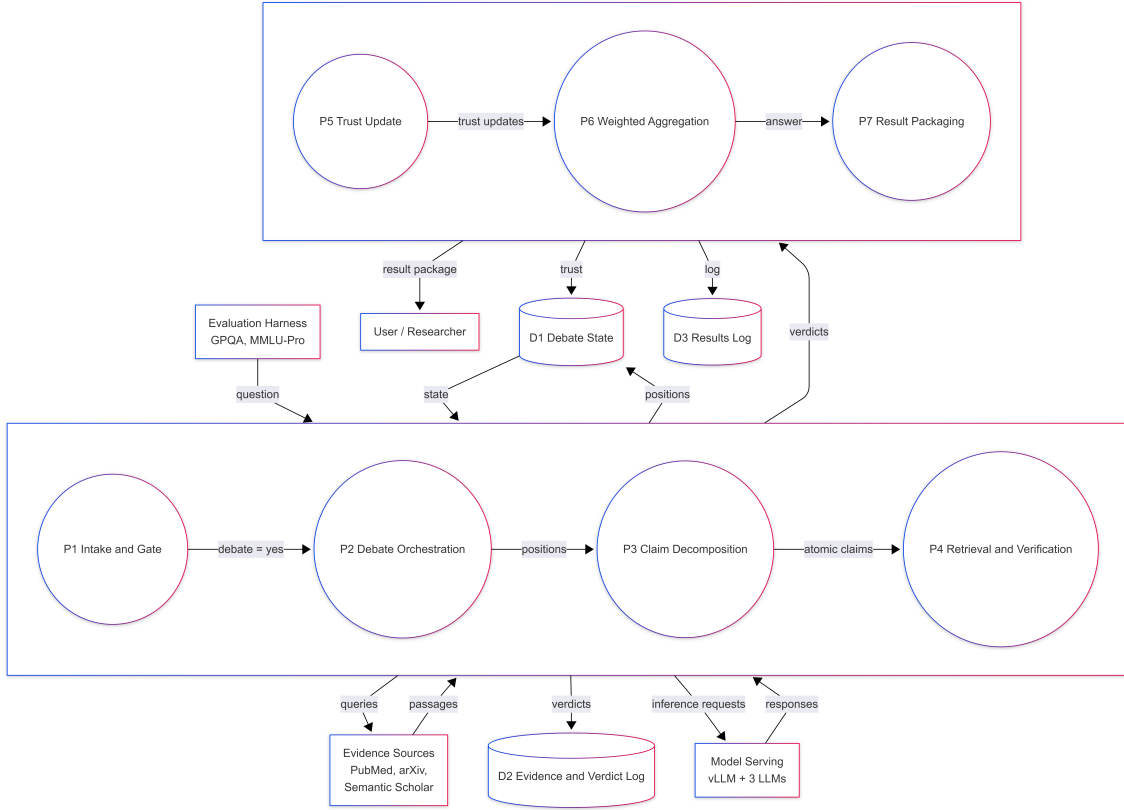


Figure 3.2: Level-one data flow diagram of the trust-calibrated multi-agent deliberation framework.

3.1.4 UI Design

The framework is operated through a command-line interface and a small web dashboard. The command-line interface starts experiments, points the pipeline at the evaluation datasets, and writes the result of every run to a JSON package. It is the primary interface because the system runs in batches on a rented GPU, where a graphical interface adds little value. The web dashboard is used afterwards to inspect single debates. It shows the agents as cards with their current positions, a chart of the trust trajectory across rounds, and the verdict of every claim with the passage that supports or contradicts it. This is where the transparency of the framework becomes visible, because a user can trace exactly why an agent lost influence. Figure 3.3 and Figure 3.4 show the intended layout of both interfaces. The screenshots will be replaced with the real output of the system once the implementation is complete.

3.2 Detailed Methodology and Design

Several design points had more than one reasonable answer, and the alternatives were compared before the current design was fixed.

The first choice is how to combine the agents at the end. Plain majority voting is

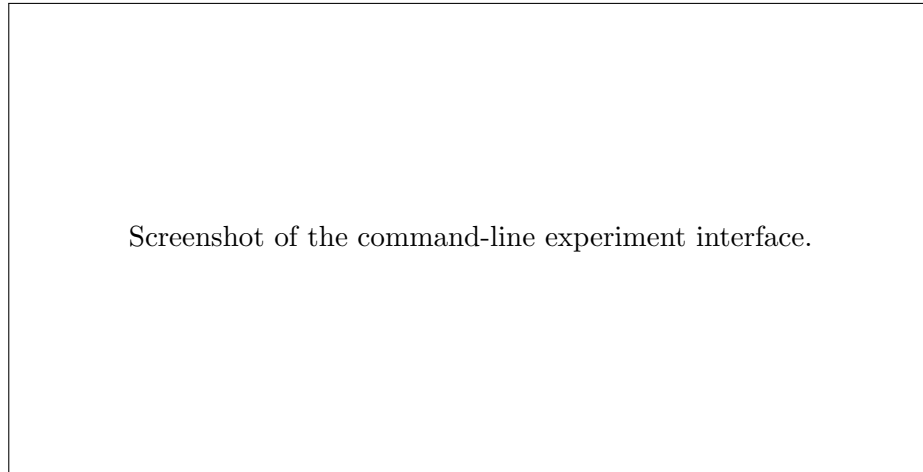


Figure 3.3: Command-line interface for running experiments and writing JSON result packages.

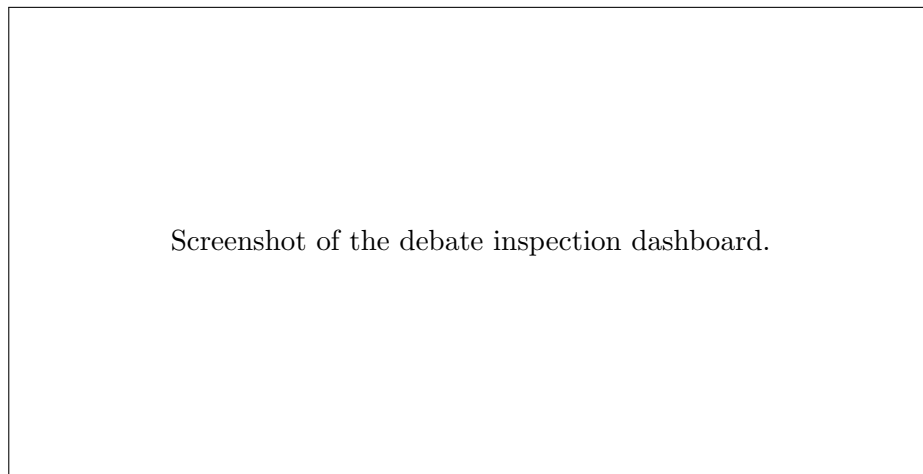


Figure 3.4: Web dashboard showing agent positions, the trust trajectory, and per-claim evidence verdicts.

the baseline the framework is measured against, so it stays in the harness but not in the framework. LLM-as-judge was rejected early. A judge shares the correlated errors of the debating models, and published results show negative net gain for that approach [8]. Trust-weighted aggregation was selected because the verdicts already exist in the pipeline, so the final answer uses the same evidence signal as the intermediate rounds.

The trust signal itself went through the same comparison. Self-reported confidence is cheap but manipulable, and its ceiling is well documented [6]. A ground-truth oracle cannot run in practice. A post-hoc behavioral classifier, as in Minority Sentinel, operates after the debate and cannot change influence during it [8]. Claim-level verification against retrieved literature was chosen because it is the only signal that comes from outside the model family.

Retrieval options also competed. A single database was simpler, but scientific claims spread across medical, physics, and general literature, so each agent searches a different

source with OpenAlex as the fallback. BM25 ranks the candidates cheaply, and the cross-encoder reranks them for semantic fit, which plain keyword scoring cannot capture.

3.3 Project Plan

3.4 Task Allocation

3.5 Summary

Chapter 4

Implementation and Results

This chapter records the environment the framework runs in and the way it is tested. Full results belong to the final report, so this chapter keeps the setup and the evaluation plan.

4.1 Environment Setup

The implementation uses Python 3.11 with vLLM serving three models through an OpenAI-compatible endpoint on a rented RTX PRO 6000 Blackwell 96GB GPU. The evidence layer calls PubMed, arXiv, Semantic Scholar, and OpenAlex over their public APIs, and every run writes a JSON result package with the trust trajectory, the verdicts, and the answer.

4.2 Testing and Evaluation

Testing follows the protocol from Chapter 3: seeded runs, the injection point for fabricated wrong consensus, and comparison against majority voting, MoA, iMAD, and DebUnc. The metric set stays fixed at accuracy, consensus-collapse rate, minority-preservation rate, and evidence-calibration rate on GPQA and MMLU-Pro.

4.3 Results and Discussion

The results will be reported once the implementation is complete. This section will present the per-dataset tables, the ablation of the trust parameters α and β , and the collapse measurements from the injection experiments.

4.4 Summary

The chapter fixes the runtime environment and the evaluation procedure. The results and their discussion arrive with the completed implementation in the final report.

Chapter 5

Standards and Design Constraints

This chapter records the standards the framework follows, the constraints that shaped its design, and the budget behind the experiments. It also maps the project against the complex engineering problem criteria and closes with a short summary.

5.1 Compliance with the Standards

Standards matter in three places: the software stack, the hardware the framework runs on, and the communication between components. Each area had alternatives, and each choice is justified below.

5.1.1 Software Standards

Python 3.11 was chosen as the implementation language. The LLM ecosystem is built around it, so vLLM, the Hugging Face libraries, and the evaluation tooling all ship Python support first. Java and C++ are faster in raw terms, but neither has the same depth of ready-made support for model serving and retrieval work. The framework is a research prototype, not a latency-critical production service, so ecosystem fit outweighs raw speed. Surveys of LLM-based multi-agent systems describe this design space in terms of workflow, infrastructure, and open challenges [15], and the choices below follow that map.

PyTorch 2.x is the deep learning framework underneath vLLM. It is the de facto standard for LLM inference and supports the Qwen, Gemma, and Mistral families directly. TensorFlow and JAX were considered. TensorFlow’s LLM serving story is weaker, and JAX would pull the whole pipeline onto its own stack without any benefit here.

vLLM serves the models through an OpenAI-compatible API. That interface matters because the framework must work with closed APIs and locally served models without touching the orchestration code. PagedAttention gives vLLM high throughput under the three-agent debate load, and its FP8 support matches the Blackwell GPU. Ollama is easier to set up but slower under concurrent requests. llama.cpp runs well on CPU but gives up the GPU throughput the debate needs. TGI is a close alternative, and vLLM won on ecosystem fit and the OpenAI-compatible surface.

Sentence-transformers provides the cross-encoder that reranks retrieved passages. BM25 was the fallback, but it matches on surface terms only, while the cross-encoder scores semantic fit between a claim and a passage. The web dashboard is served with FastAPI, and every result package is written as JSON following RFC 8259, validated against a JSON Schema. Git tracks the code and LaTeX (Overleaf) produces this report.

5.1.2 Hardware Standards

The experiments run on a single NVIDIA RTX PRO 6000 Blackwell GPU with 96 GB of GDDR7 memory. The choice came down to memory capacity and architecture. The development models (Qwen3.5-9B, Gemma 4 12B, Ministral-3-14B) fit comfortably in BF16. The final models (Qwen3.6-27B, Gemma 4 26B, Mistral Small 3.2 24B) total about 77 billion parameters, which no single card holds in FP16, so they run in FP8. Blackwell has native FP8 support, and that is exactly the point of difference.

The A100 80GB was the first alternative considered. It has enough memory for the development models and strong HBM bandwidth, but its FP8 path is weaker and the architecture is two generations old. The H100 80GB is faster yet rents for roughly twice the price and brings nothing the framework needs. The L40S at 48 GB and the consumer RTX 5090 at 32 GB cannot hold the final model trio at all. The RTX PRO 6000 sits at the right point: enough memory, modern quantization, and a rental price that fits a student budget. The host machine needs at least 64 GB of system RAM and NVMe storage to stage the models before they move to the GPU.

5.1.3 Communication Standards

All communication between the framework and its components runs over HTTP with TLS. The model server exposes the OpenAI-compatible chat completions endpoint, which is REST with JSON payloads. gRPC was the alternative for model serving. It is faster in benchmark terms, but the OpenAI-compatible surface keeps the framework portable between closed APIs and local vLLM servers, and that portability outweighs the latency difference.

The evidence sources are reached through their public REST interfaces: PubMed E-utilities, the arXiv API, the Semantic Scholar Graph API, and OpenAlex as the fallback. Each returns JSON or XML over HTTPS, and the framework handles rate limits with the retry caps defined in the requirements. WebSocket is reserved for the dashboard’s live trust chart. Nothing else needs a persistent connection.

5.2 Design Constraints

5.2.1 Economic Constraint

The budget is the tightest constraint. Everything runs on a single rented GPU with no fine-tuning, because training, or even parameter-efficient tuning, would multiply the cost

of every experiment. The RTX PRO 6000 rental range of about USD 0.70 to 1.90 per hour keeps the total within reach of a student budget, and the retrieval APIs used are free under their rate limits. Owning the card was never an option; its retail price is several times the whole project budget.

5.2.2 Environmental Constraint

The GPU draws up to about 600 watts under load. Running it in the cloud means the hardware is shared across many tenants, so one project does not manufacture or discard a card. Skipping fine-tuning also keeps energy use down, because inference over a few thousand questions costs far less than a training run that would occupy the same card for days.

5.2.3 Ethical Constraint

The framework keeps the ground truth hidden from every agent, and the final output carries citations to the evidence behind the answer. Models tuned to sound agreeable become less accurate and more sycophantic [4], and preference-based sycophancy fixes need careful data handling [22]. In multi-agent debate, the correct answer can be present but ignored in more than 20 percent of wrong-answer cases [3]. Making the evidence trail visible is the framework’s answer to those risks. A user can always check which agent was supported and which was contradicted.

5.2.4 Health and Safety Constraint

The work involves no physical hardware handling and no laboratory risk. All compute runs on rented cloud instances. The only health consideration is ordinary screen time, which the team manages with standard breaks and workstation habits.

5.2.5 Social Constraint

Scientific question answering shapes how people come to trust AI systems. A system that confidently repeats a wrong consensus can reinforce misinformation [19]. The framework grounds influence in evidence, so the social risk is reduced to the quality of the retrieved literature rather than the popularity of an answer.

5.2.6 Political Constraint

The framework relies on open access to scientific literature. PubMed, arXiv, Semantic Scholar, and OpenAlex are all public services, and the models are open-weight with permissive licenses: Qwen and Mistral under Apache 2.0, Gemma under its own license. No user data is collected, and the system depends on no single vendor, which keeps it usable under changing access policies.

5.2.7 Sustainability

Sustainability here comes from reuse. The rented GPU is shared infrastructure, the models are already trained and only served, and every experiment is seeded and logged so runs do not need to be repeated. Open licenses and reproducible result packages mean the work can be continued by others without redoing the setup.

5.3 Cost Analysis

The budget covers the rented GPU, storage, and the free research services. The largest line is inference time on the RTX PRO 6000 Blackwell, estimated at about 300 hours across development and evaluation. Table 5.1 summarizes the primary budget.

Table 5.1: Primary budget for the project.

Item	Quantity	Unit Cost (USD)	Total (USD)
RTX PRO 6000 Blackwell 96GB rental	300 hours	0.70–1.90 / hour	210–570
Retrieval APIs (PubMed, arXiv, S2, OpenAlex)	—	0 (free tier)	0
Datasets (GPQA, MMLU-Pro)	—	0 (public)	0
Storage (model weights, logs)	200 GB	0.05 / GB	10
Total			220–580

An alternate budget was also prepared. Renting an A100 80GB instead costs about USD 0.68 to 1.50 per hour, or 204–450 for the same 300 hours. The saving is real but small, and it comes at a cost: weaker FP8 support means the final model trio would need tighter quantization or sequential loading, which adds engineering time. The RTX PRO 6000 was selected because the Blackwell FP8 path keeps the pipeline simple.

The project has no revenue model. It is an academic deliverable, funded by the university and the team, and its outputs are the report and the open-source framework. Table 5.2 shows the alternate budget for comparison.

Table 5.2: Alternate budget using the A100 80GB.

Item	Quantity	Unit Cost (USD)	Total (USD)
A100 80GB rental	300 hours	0.68–1.50 / hour	204–450
Retrieval APIs	—	0 (free tier)	0
Datasets	—	0 (public)	0
Storage	200 GB	0.05 / GB	10
Total			214–460

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

The project qualifies as a complex engineering problem under the Washington Accord criteria. Table 5.3 maps the seven attributes P1–P7, and the subsections that follow give the rationale for each mapping.

Table 5.3: Mapping with complex problem solving.

P1 Dept of Knowledge	P2 Range of Conflicting Requirements	P3 Depth of Analysis	P4 Familiarity of Issues	P5 Extent of Applicable Codes	P6 Extent of Stakeholder Involvement	P7 Inter-dependence
✓	✓	✓	✓	✓	✓	✓

P1: Depth of Knowledge Required

The framework spans machine learning, natural language processing, retrieval-augmented generation, and distributed model serving. Solving it required specialist knowledge in all four areas, from the trust update formula to the vLLM serving layer. Table 5.4 maps the P1 knowledge profile.

The knowledge profile is broad rather than deep in one spot. The report applies DK4 and DK5 in every chapter, and DK6 and DK7 in the implementation, which reflects the range the problem demands.

Table 5.4: Knowledge profile mapping for P1.

Knowledge	Applied	Where used
DK1: Natural sciences	✓	Statistics behind trust updates and metrics
DK2: Mathematics	✓	Softmax, clamping, renormalization, probability
DK3: Engineering fundamentals	✓	System design, state machines, failure handling
DK4: Specialist knowledge	✓	LLMs, multi-agent debate, RAG, sycophancy
DK5: Engineering methods	✓	Experimental methodology, baseline comparison
DK6: Computational methods	✓	Python, PyTorch, vLLM, FastAPI
DK7: Codes and practices	✓	JSON, HTTP, TLS, version control
DK8: Social context	✓	Ethical constraints on AI output

P2: Range of Conflicting Requirements

Accuracy competes with latency and cost. More debate rounds improve the signal but multiply inference time and the GPU bill. Trust calibration competes with diversity, because strong trust weighting can silence a minority that the evidence simply has not covered yet. Evidence coverage competes with API rate limits. These conflicts had to be resolved jointly, and the round caps, the clamping bounds, and the conservative verdict handling are the result.

P3: Depth of Analysis

The problem has no textbook solution. Answering it required a comparative experiment with multiple baselines (majority voting, MoA, iMAD, DebUnc) and a metric set designed for the failure mode: consensus-collapse rate, minority-preservation rate, and evidence-calibration rate, alongside plain accuracy. The injection point for fabricated wrong consensus was built specifically so the collapse dynamic could be measured under controlled pressure.

P4: Familiarity of Issues

Sycophantic consensus in multi-agent LLM debate is a new research area. The closest published work appeared between 2024 and 2026, and no standard solution exists. The

team started with a full literature review, which is documented in Chapter 2, before committing to a design.

P5: Extent of Applicable Codes

General software standards apply: JSON (RFC 8259), HTTP over TLS, the OpenAI-compatible API, and JSON Schema validation. No standard, however, covers evidence-grounded trust in LLM debate. The problem is therefore only partially standardized, and the framework defines its own conventions for the trust score and the verdict set.

P6: Extent of Stakeholder Involvement

The stakeholders are the supervisor, the examiners, and the users of scientific question-answering tools. The supervisor guided the scope and the evaluation plan, and the examiners assess the report against the complex engineering criteria. Future users matter because the dashboard and the result package were designed around traceability, not just accuracy.

P7: Interdependence

The confidence gate, the orchestrator, the claim decomposer, the retrieval subsystem, the trust updater, and the aggregator are tightly coupled. A failure in retrieval cascades into the verdicts, then into trust, then into the final answer. The failure-isolation requirement in Section 3.1.1 exists because of this coupling.

5.4.2 Engineering Activities

Table 5.5 maps the project to the five engineering activities A1–A5, with rationales below.

Table 5.5: Mapping with complex engineering activities.

A1 Range of re- sources	A2 Level of Inter- action	A3 Innovation	A4 Consequences for society and environ- ment	A5 Familiarity
✓	✓	✓	✓	✓

A1: Range of Resources

The project draws on three model families, four literature APIs, a rented Blackwell GPU, and six team members with different strengths. Coordinating those resources is part of

the engineering work, and the project plan in Chapter 3 assigns each task to a named member.

A2: Level of Interaction

The system interacts heavily with the outside world. Every debate round calls three external model endpoints and three literature APIs, and the cloud GPU is reached over the network. The team also interacts through a shared repository and weekly reviews.

A3: Innovation

The trust-calibrated aggregation mechanism is new. No published method ties an agent’s influence to claim-level verification against retrieved scientific evidence while the debate is still running. The gap analysis in Chapter 2 documents this, and the design in Chapter 3 implements it.

A4: Consequences for Society and Environment

The social consequence is improved reliability in AI scientific question answering, which lowers the risk of confidently wrong answers being trusted. The environmental consequence is modest, since the work runs on shared cloud hardware without fine-tuning.

A5: Familiarity

Multi-agent debate and evidence verification were new territory for the team. The literature review phase, the baseline study, and the incremental implementation plan were all designed around that reality, so each module was built only after its background was understood.

5.5 Summary

The chapter fixed the standards behind the software, the hardware, and the communication layers, and recorded the economic, environmental, ethical, and social constraints that shaped the design. The budget analysis shows the project fits a student budget on a rented RTX PRO 6000 Blackwell GPU, and the complex engineering mapping confirms the project meets all seven problem attributes and all five engineering activities.

Chapter 6

Conclusion

This chapter closes the report with a short summary of the work, its known limitations, and the directions the team sees for follow-up work.

6.1 Summary

The report proposed a trust-calibrated multi-agent deliberation framework that replaces majority voting with evidence-grounded trust weighting. The design, the evaluation protocol, and the project plan are complete, and the final report will add the measured results.

6.2 Limitation

The known limits are retrieval sparsity on niche scientific topics, imperfect claim decomposition, and correlated reasoning errors between agents that share pretraining data.

6.3 Future Work

Follow-up work could test the framework on open-ended research questions, replace the cross-encoder with process reward models, and extend the debate beyond three agents.

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